Sexto Laboratorio

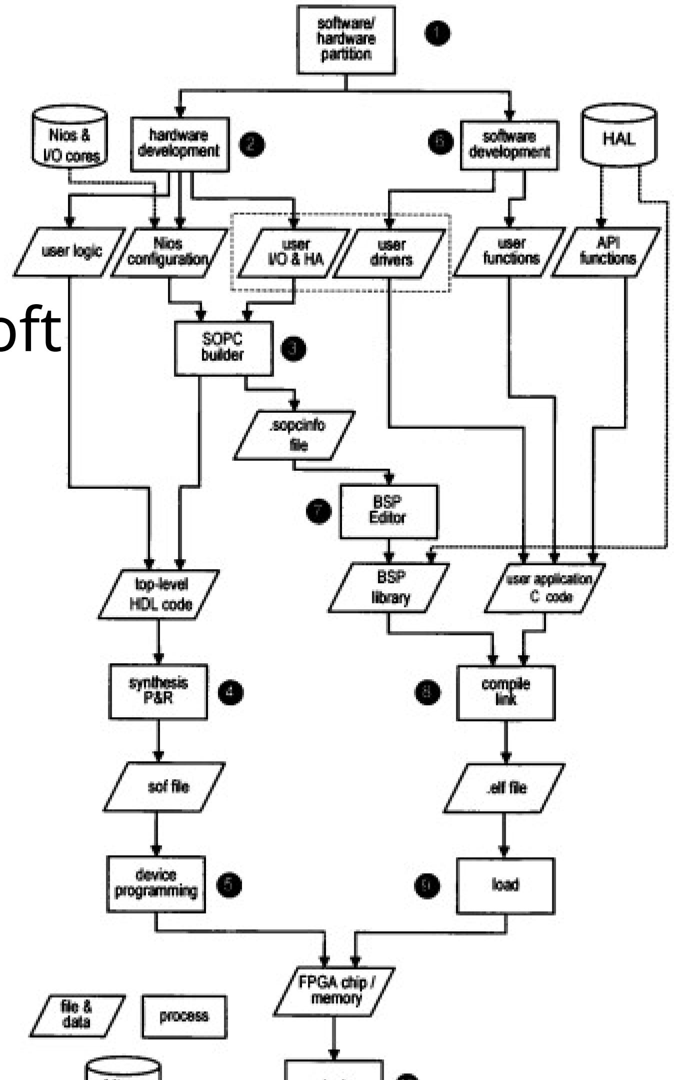
- Integración de módulos de hardware programable
- Uso de verilog
- Systembuilder/Quartus

- Desarrollo de un sistema con módulos de hardware programable

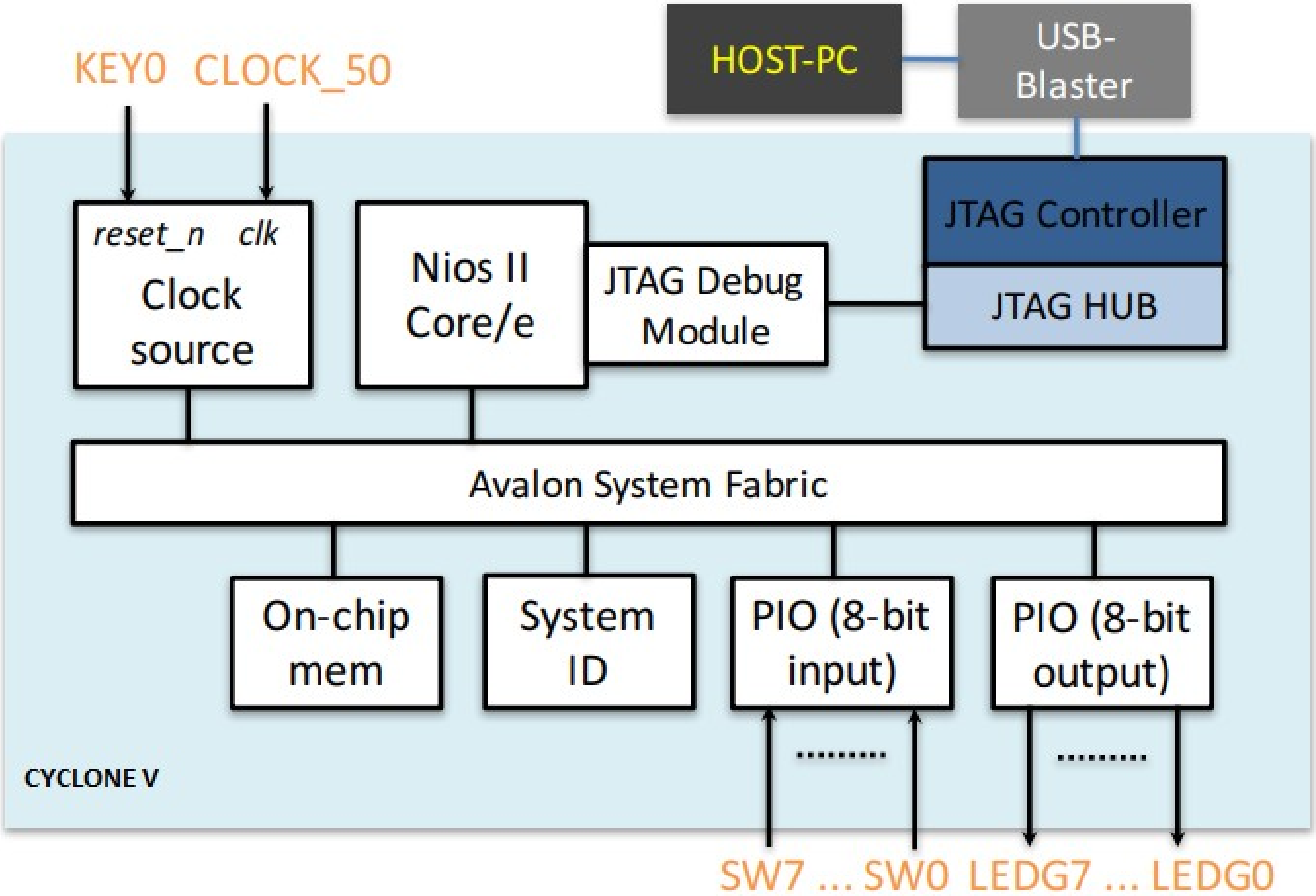
- **Construcción un SoPC**

Construcción de un SoPC

- Particionar tareas
- Desarrollar el hardware e integrarlo con el procesador soft (NIOSII)
- Desarrollar el software
- Implementar el software y el hardware y realizar el testing



Primer SoPC

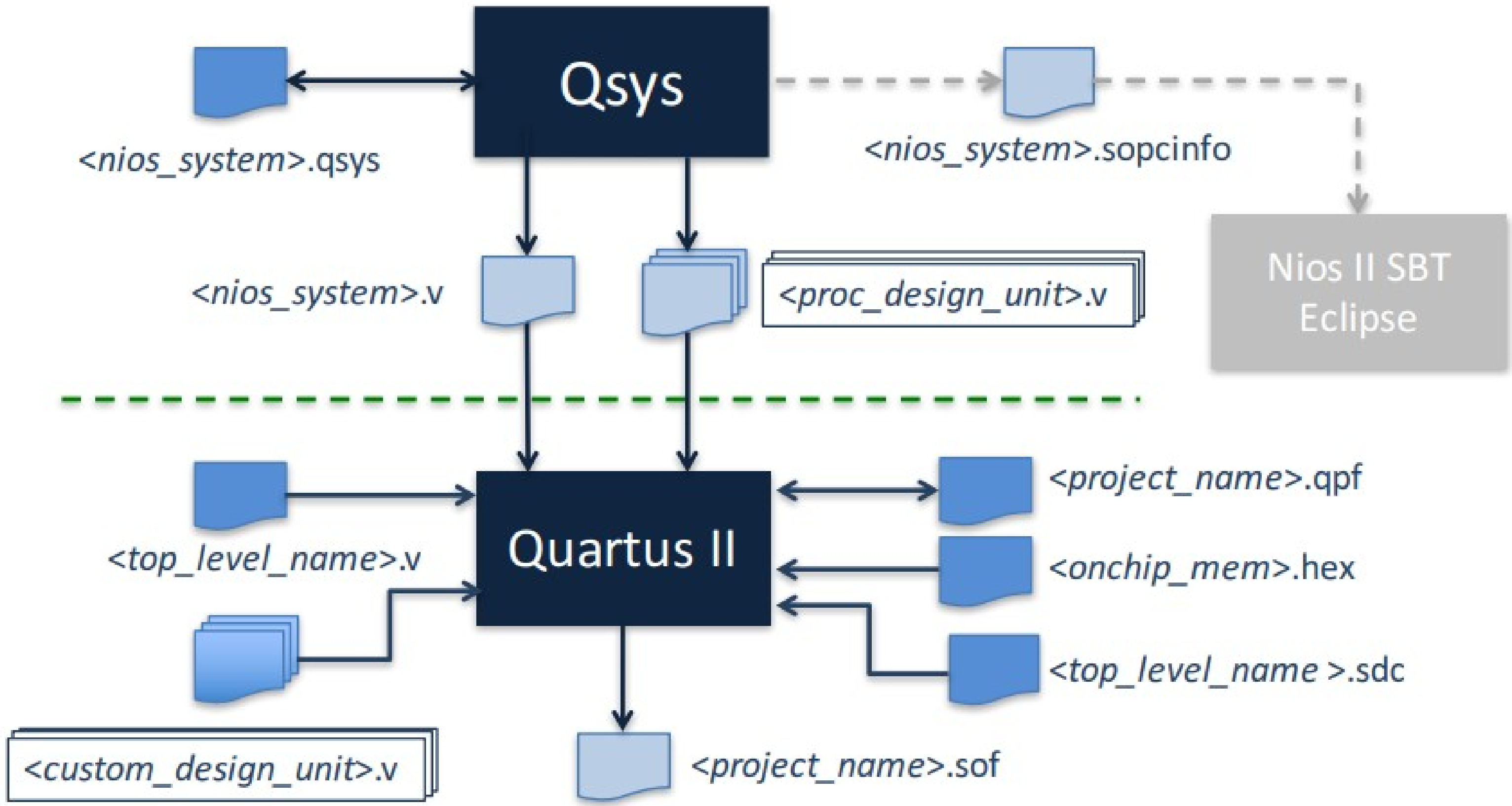


KEY0
CLOCK_50
HOST-PC
USB-
Blaster
JTAG Controller
JTAG HUB
reset_n
clk
Clock
source
Nios II
Core/e
JTAG Debug
Module
Avalon System Fabric
On-chip
mem
System
ID
PIO (8-bit
input)
PIO (8-bit
output)
....
....
CYCLONE V
SW7 ... SW0
LEDG7 ... LEDG0

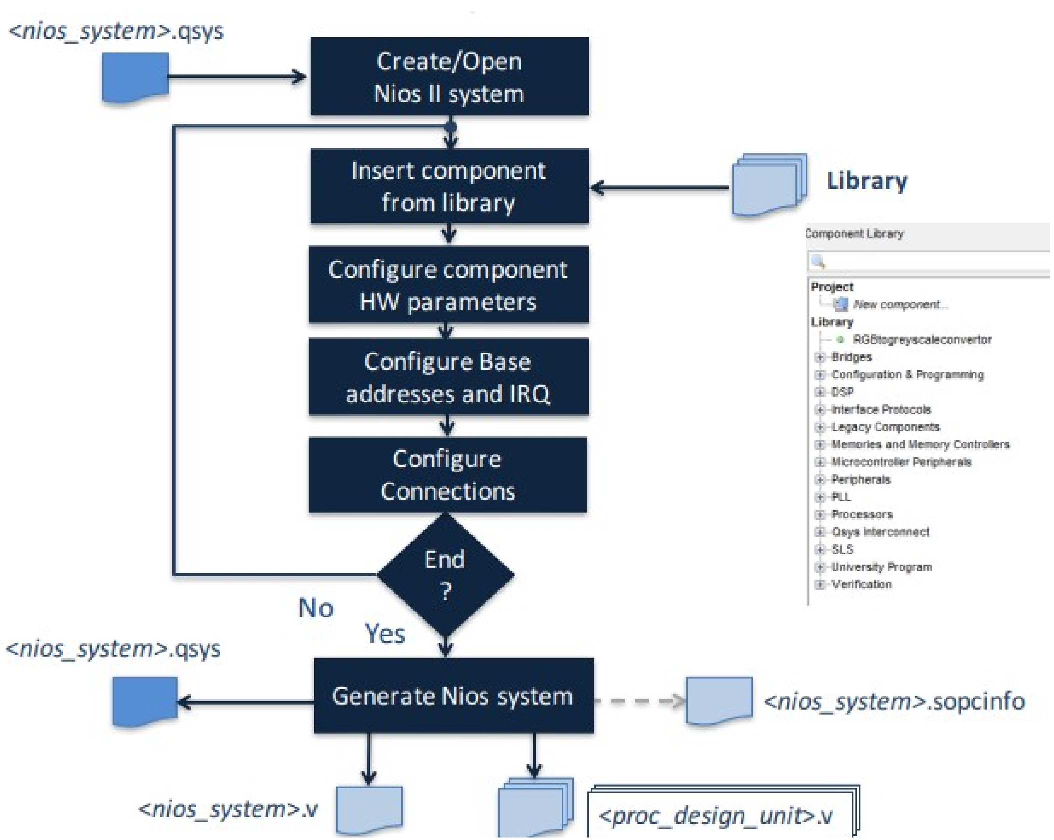
Componentes del SoPC

- Cpu NiosII (la version más económica) más JTAG Debug
- Onchip memoria para programa y datos (20K)
- 2 Puertos I/O
  - Input para leer los switches
  - Output para leer los leds
- System Id para identificación

Platform Designer (Ex-Qsys)



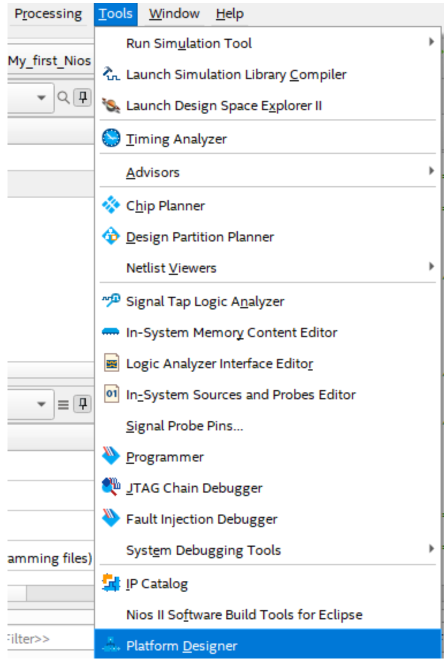
Platform Designer (Ex-Qsys)



Creación System Hardware

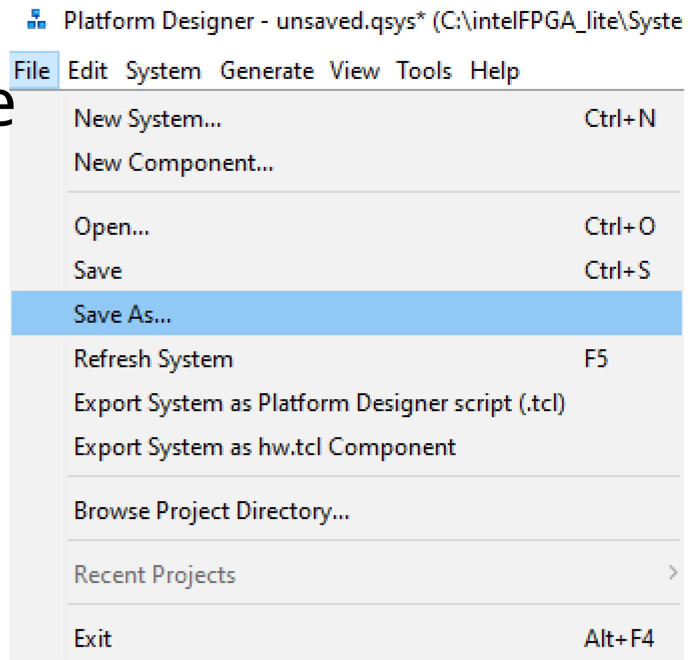
- Abrir Quartus después de crear el template con SystemBuilder

- Ir a Tools-> Platform Designer

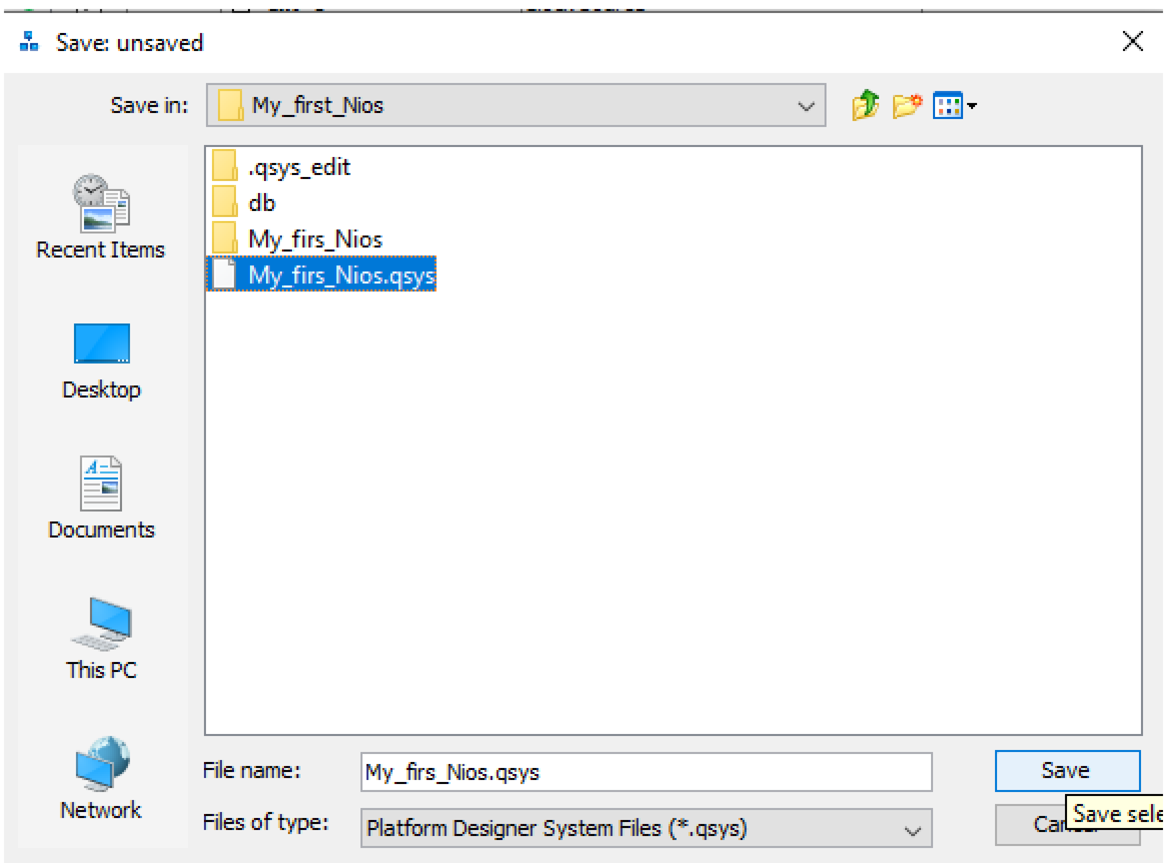


Creación System Hardware

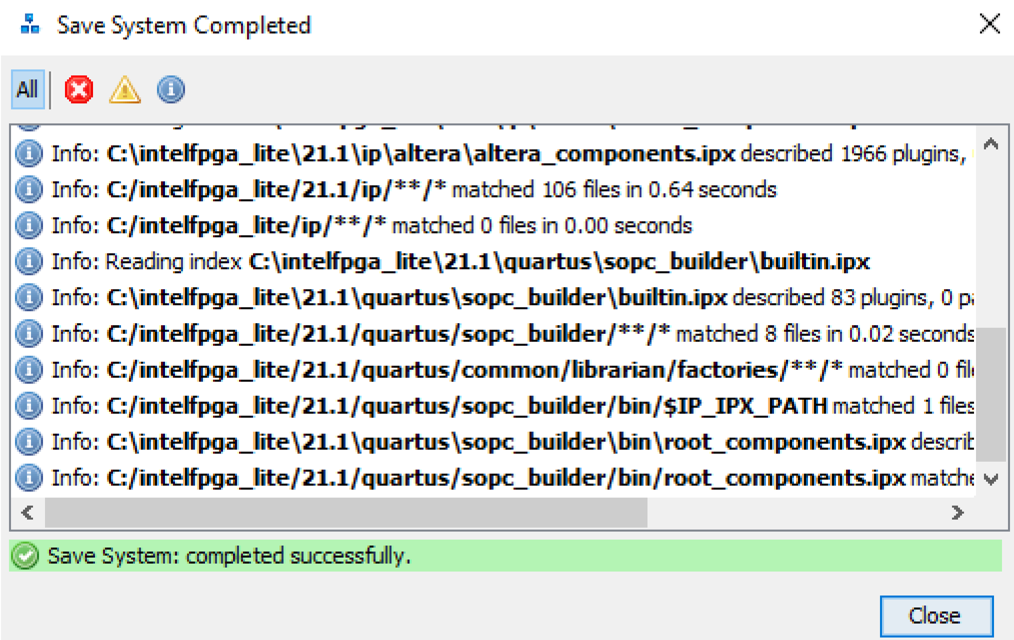
- Guardar el archivo .qsys que vamos a crear



Creación System Hardware

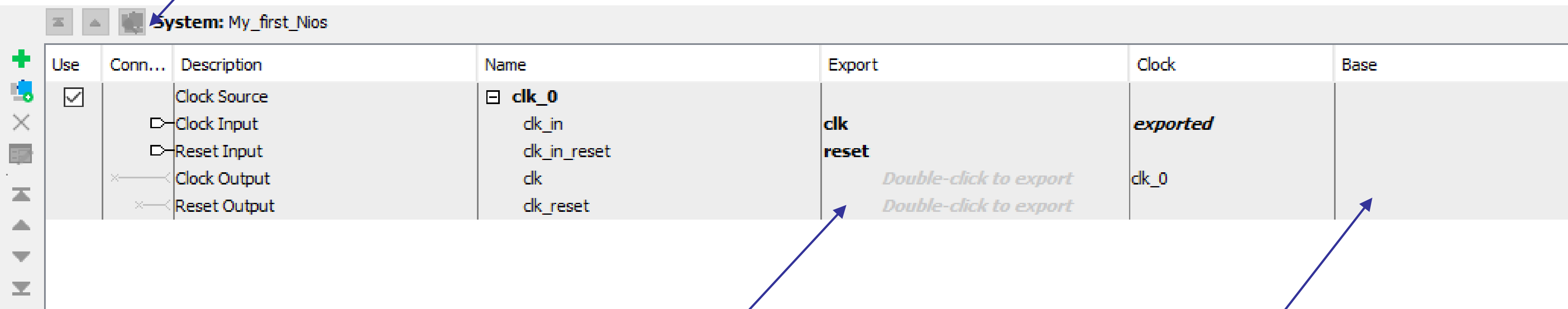


Creación System Hardware



Creación System Hardware

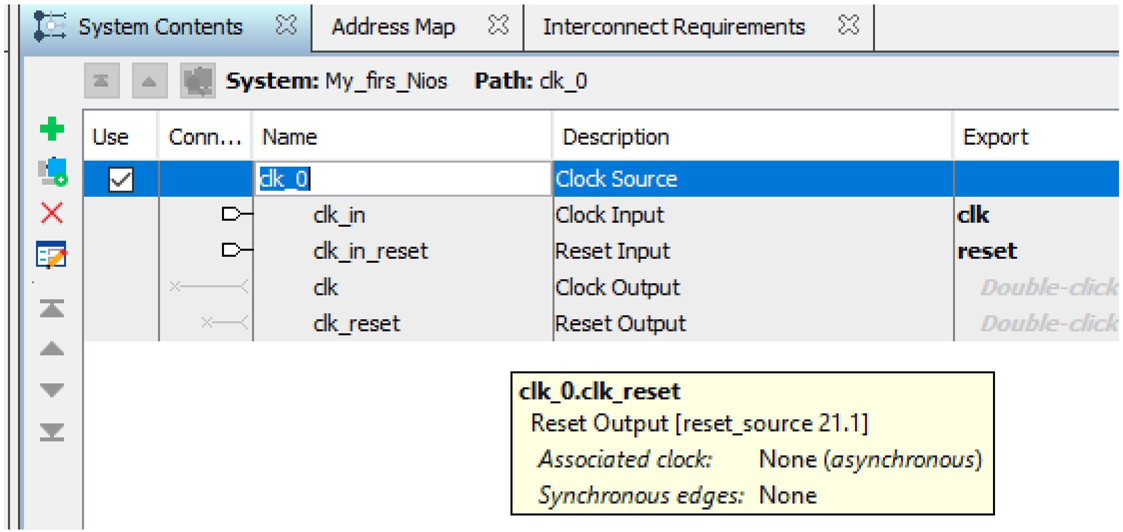
Configuración de conexiones internas



Variables que serán exportadas

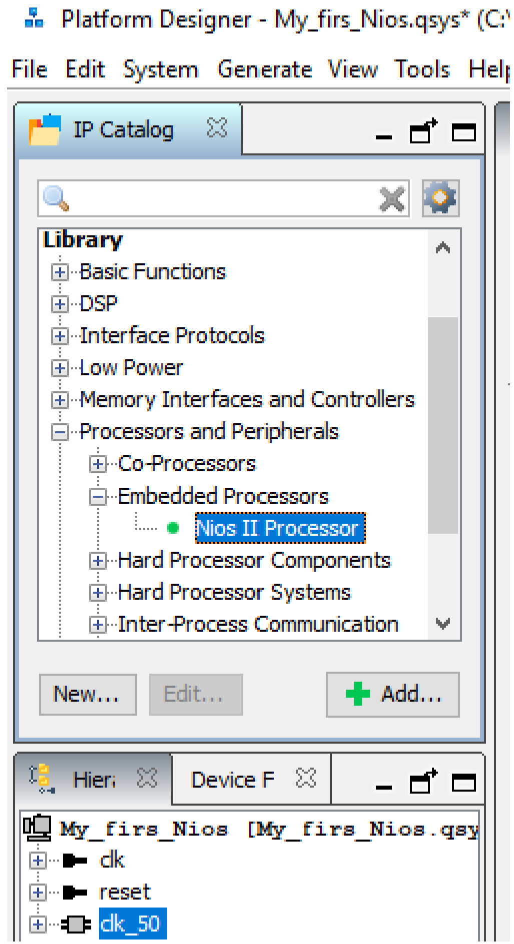
Direcciones base del sistema

Creación System Hardware

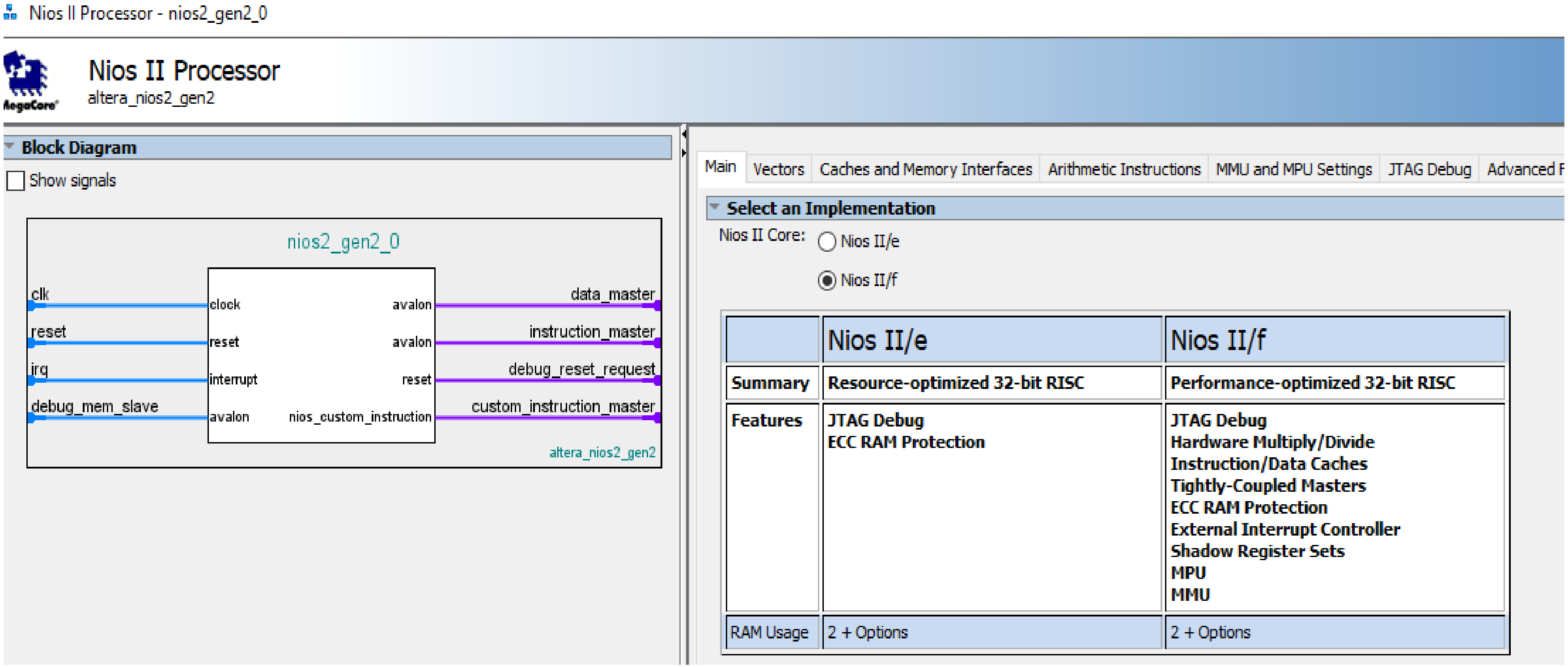


Creación System Hardware

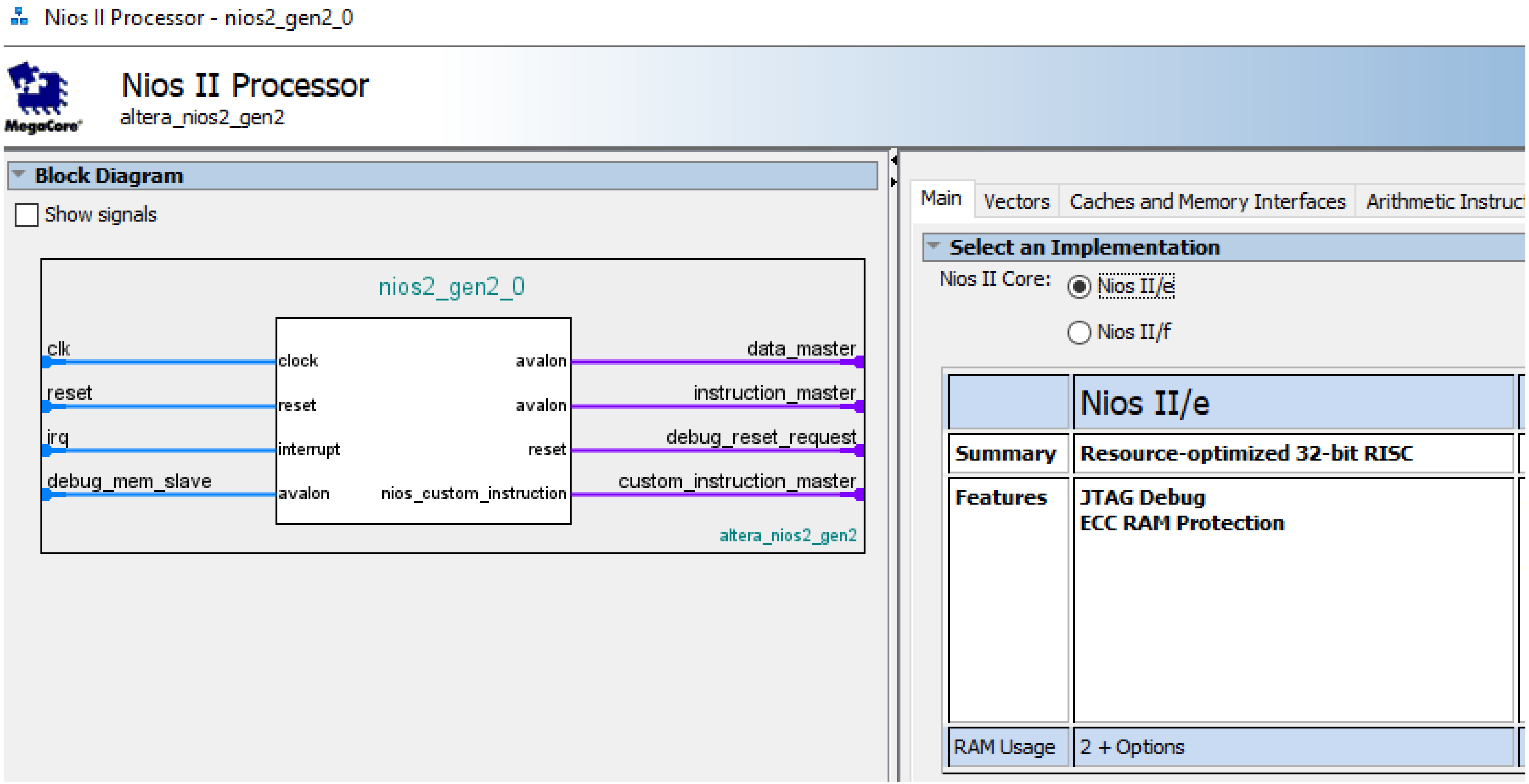
- Sumar un Nios II Processor, versión económica



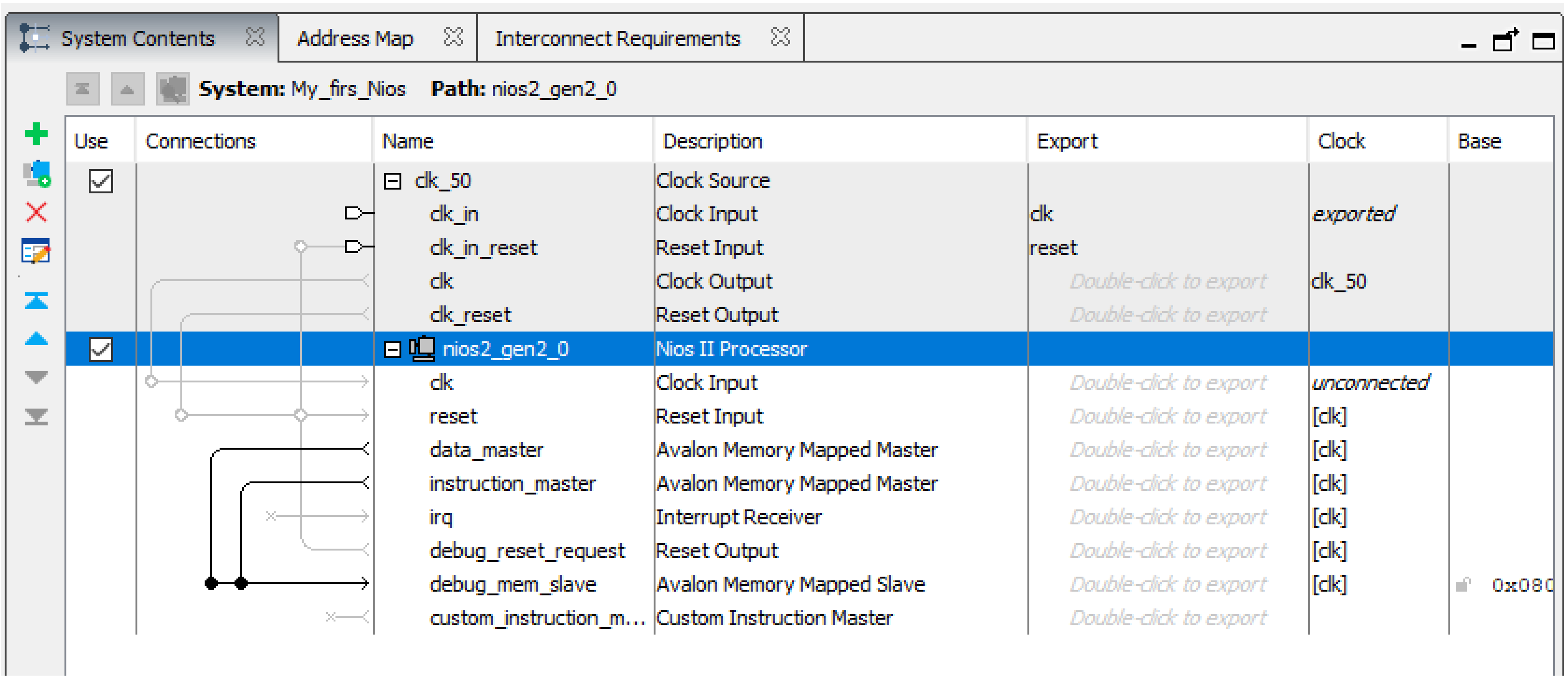
Creación System Hardware



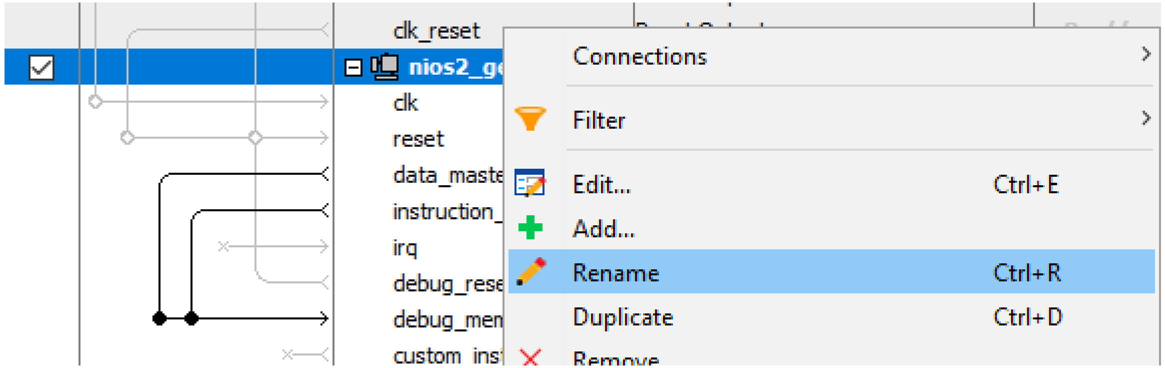
Creación System Hardware



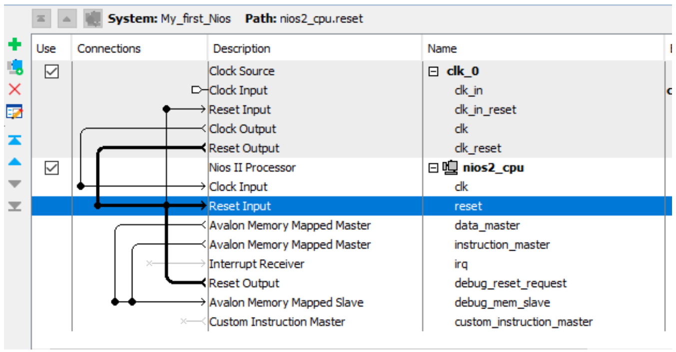
Creación System Hardware



Creación System Hardware



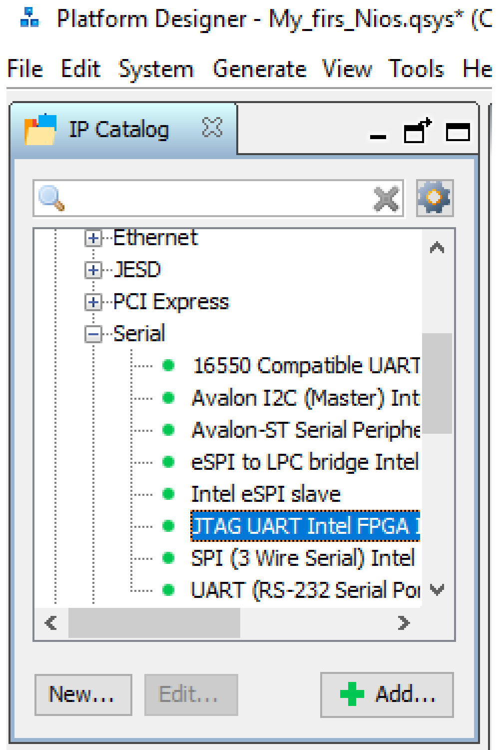
Renombrar los componentes, por ej; el nios2_gen2_0 -> nios2



Rutear el clock a los distintos componentes del sistema

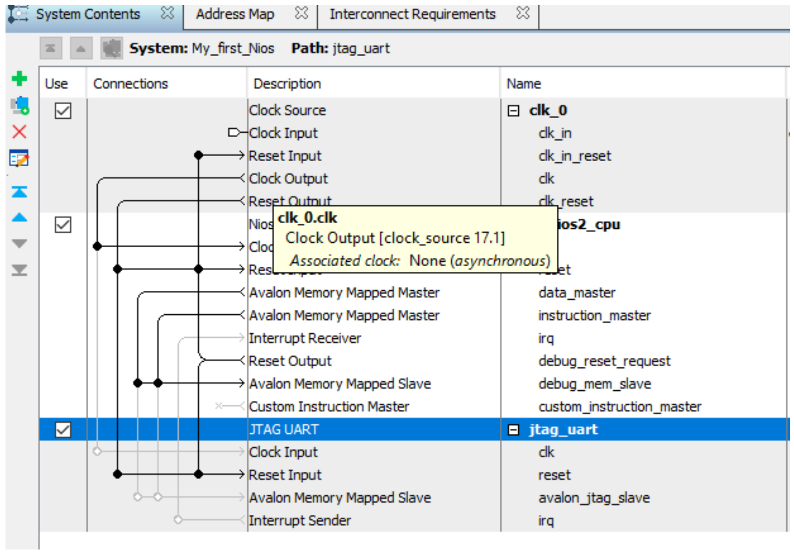
Creación System Hardware

- Sumar un JTAG UART



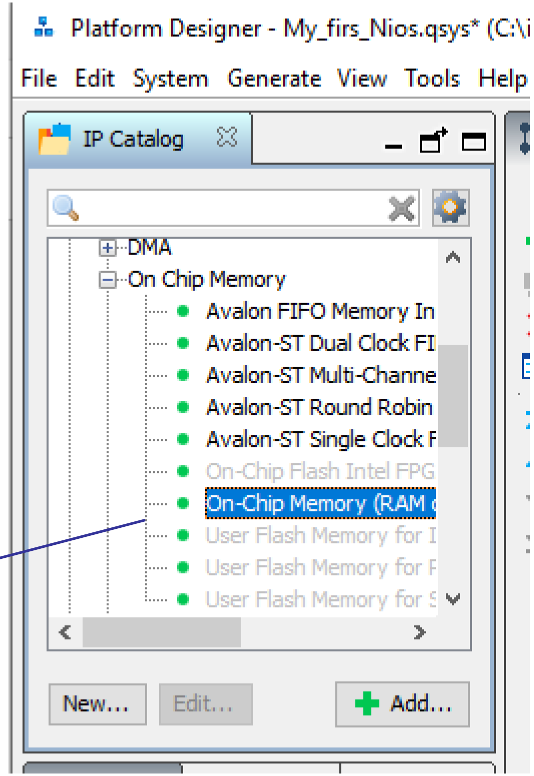
Creación System Hardware

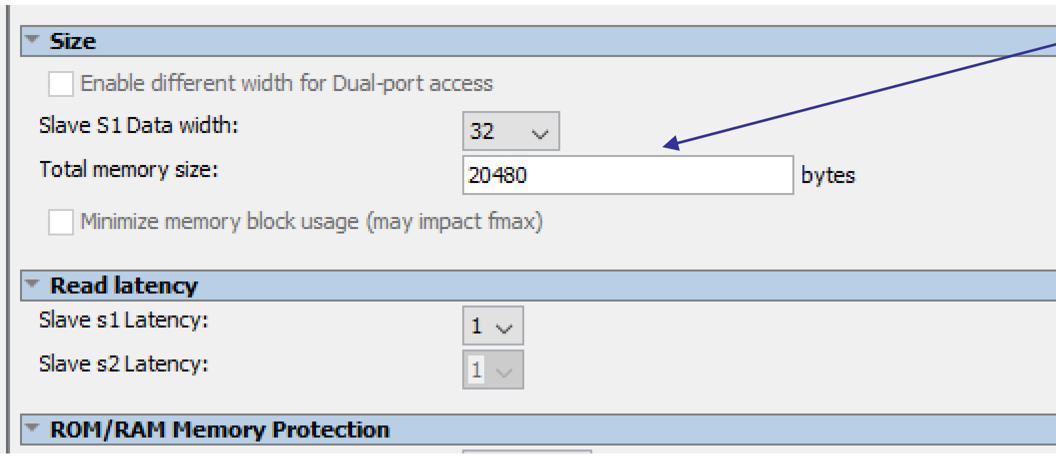
- Unir las señales reset desde el CLOCK al módulo JTAG



Creación System Hardware

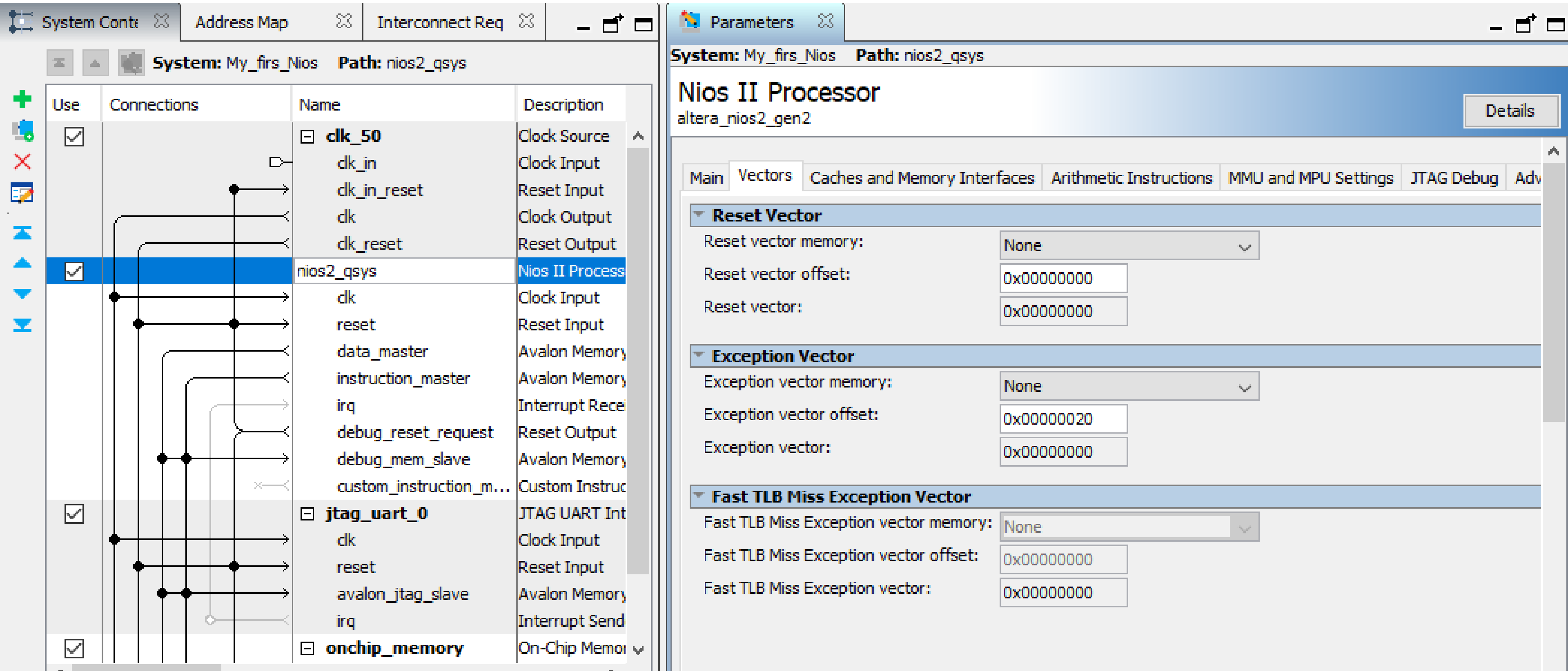
- Sumar módulo de memoria (Size 20k)



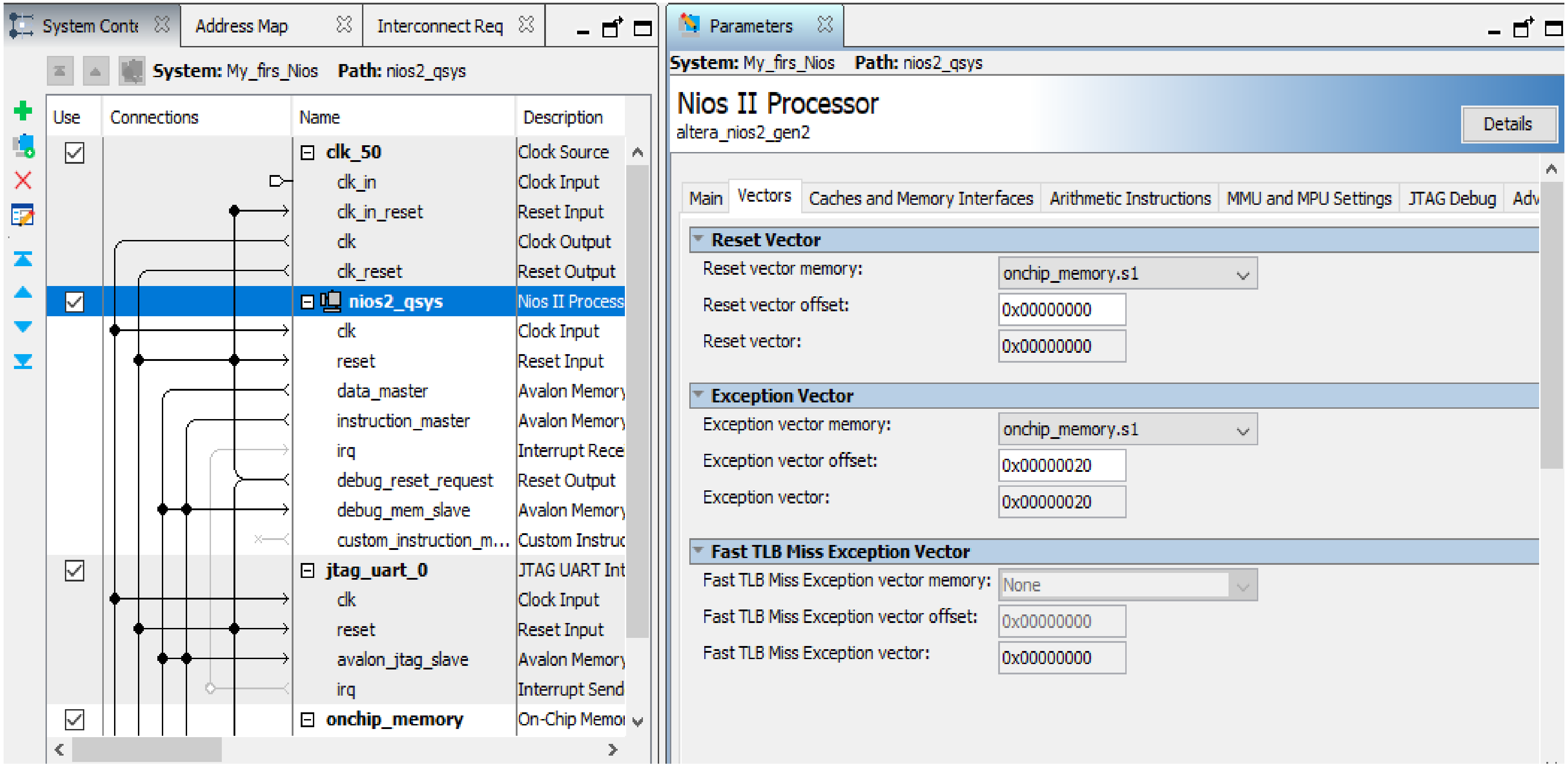


Creación System Hardware

Asignar el "Reset Vector" y "Exception Vector del Nios a la memoria"

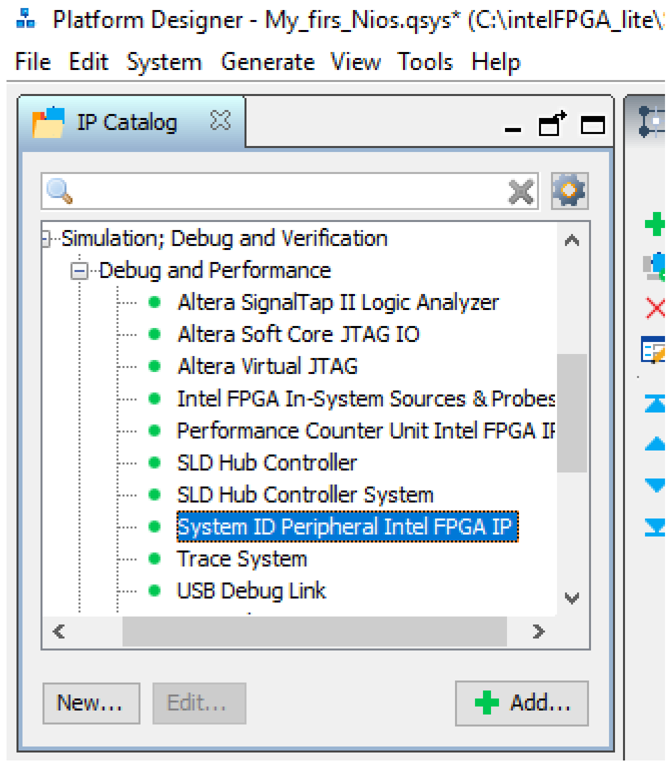


Creación System Hardware

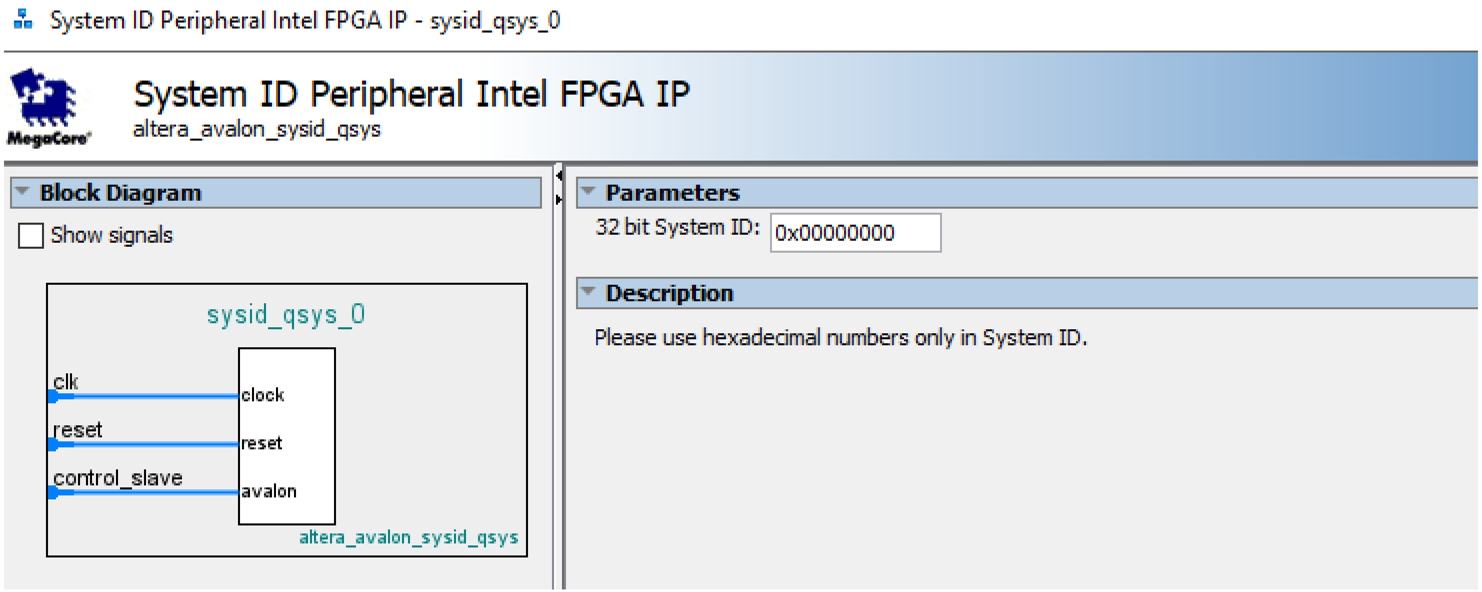


Creación System Hardware

- Sumar el periférico System ID

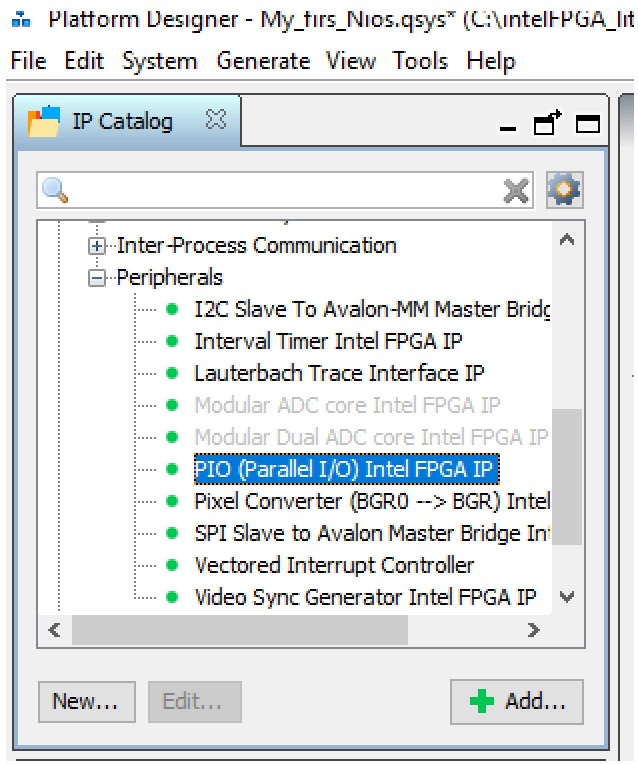


Creación System Hardware



Creación System Hardware

- Sumar el puerto de salida (renombrarlo como “leds”)
- Sumar el puerto de entrada (renombrarlo como “switches”)



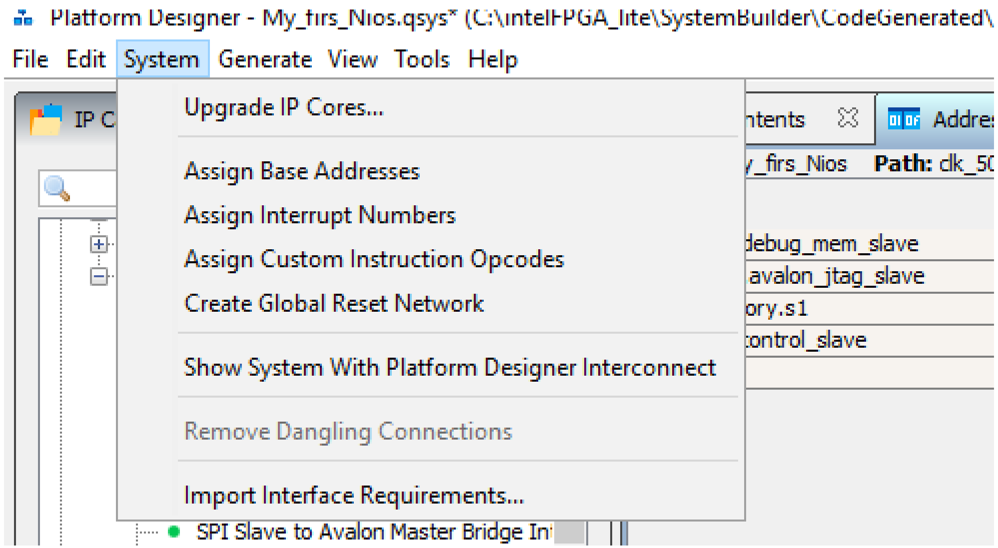
Ajustar mapa de memoria

Address Map | Interconnect Requirements

System: My_first_Nios **Path:** switches.clk

	nios2_cpu.data_master	nios2_cpu.instruction_master
nios2_cpu.debug_mem_slave	0x0800 - 0x0fff	0x0800 - 0x0fff
jtag_uart.avalon_jtag_slave	0x0000 - 0x0007	0x0000 - 0x0007
onchip_memory.s1	0x0000 - 0x4fff	0x0000 - 0x4fff
sysid_qsys.control_slave	0x0000 - 0x0007	0x0000 - 0x0007
leds.s1	0x0000 - 0x000f	0x0000 - 0x000f
switches.s1	0x0000 - 0x000f	0x0000 - 0x000f

Creación System Hardware



Creación System Hardware

Messages

Type	Path	Message
⊟ ⚠	2 Warnings	
⚠	**My_first_Nios.leds**	**leds.external_connection** must be exported, or connected to a matching conduit.
⚠	**My_first_Nios.switches**	**switches.external_connection** must be exported, or connected to a matching conduit.
⊟ ⓘ	4 Info Messages	
ⓘ	**My_first_Nios.jtag_uart**	JTAG UART IP input clock need to be at least double (2x) the operating frequency of JTAG TCK on board
ⓘ	**My_first_Nios.sysid_qsys_0**	System ID is not assigned automatically. Edit the System ID parameter to provide a unique ID
ⓘ	**My_first_Nios.sysid_qsys_0**	Time stamp will be automatically updated when this component is generated.
ⓘ	**My_first_Nios.switches**	PIO inputs are not hardwired in test bench. Undefined values will be read from PIO inputs during simulation.

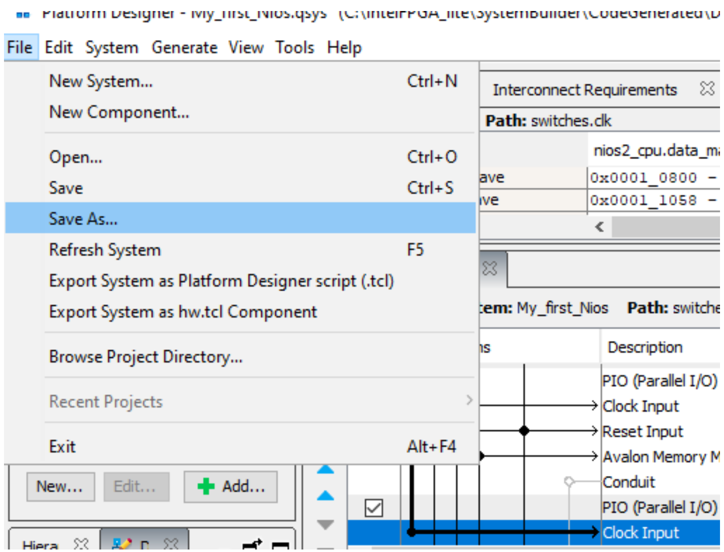
switches.external_connection must be exported, or connected to a matching condu

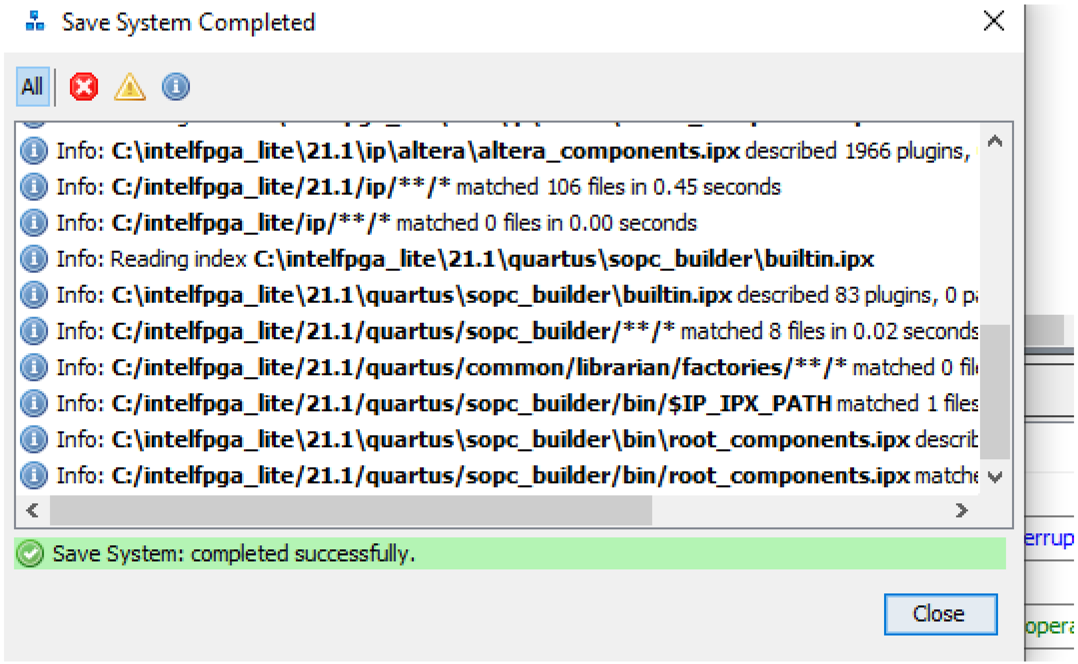
Exportar leds y switches

Name	Export	Clock	Base	End
⊟ **leds**				
clk	*Double-click to export*	**clk_0**		
reset	*Double-click to export*	[clk]		
s1	*Double-click to export*	[clk]	0x0001_1030	0x0001
external_connection	**leds_external_connection**			
⊟ **switches**				
clk	*Double-click to export*	**clk_0**		
reset	*Double-click to export*	[clk]		
s1	*Double-click to export*	[clk]	0x0001_1020	0x0001
external_connection	**switches_external_connection**			

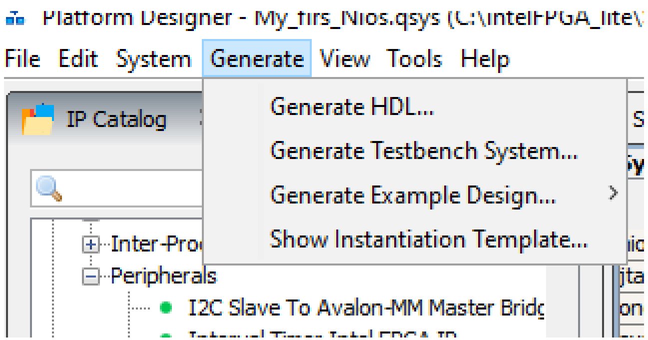
Creación System Hardware

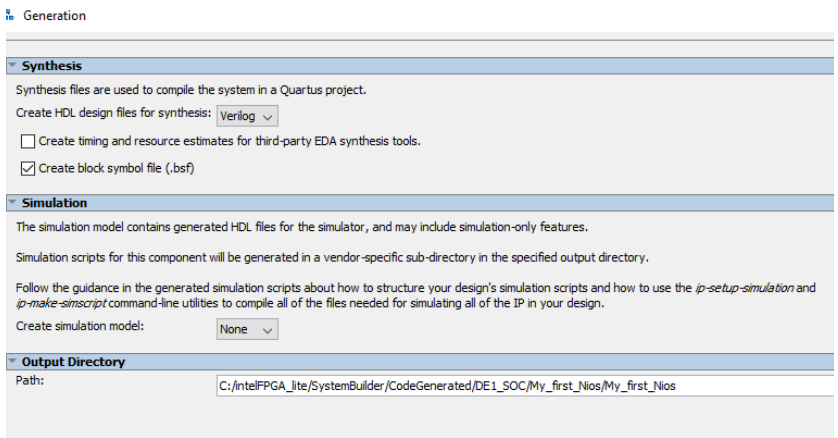
Salvar el .qsys y generar el .sopcinfo



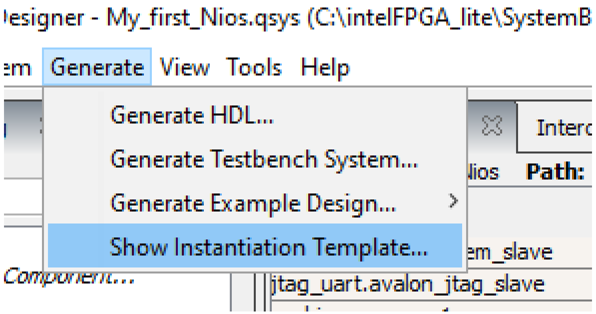


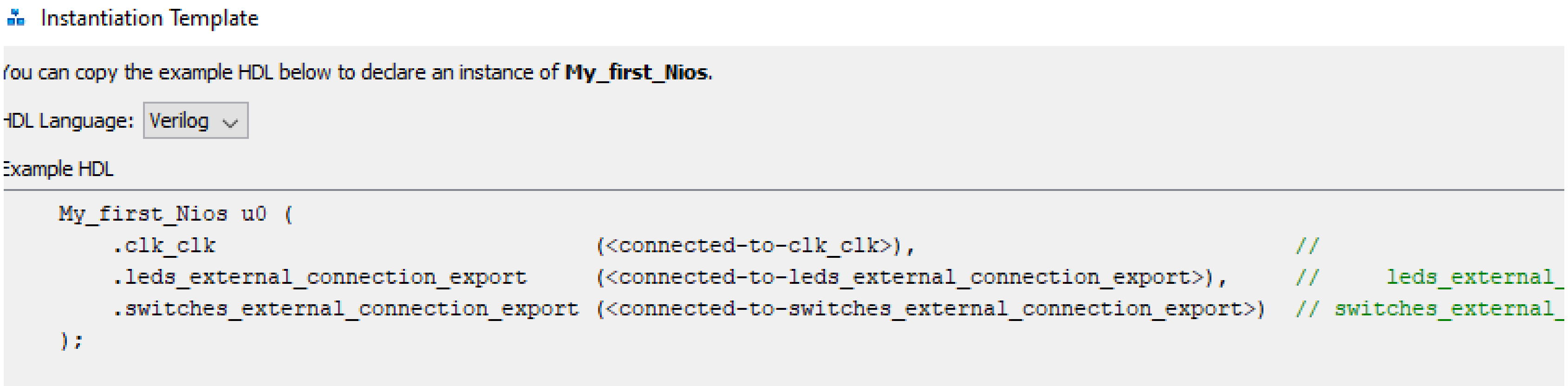
Creación System Hardware





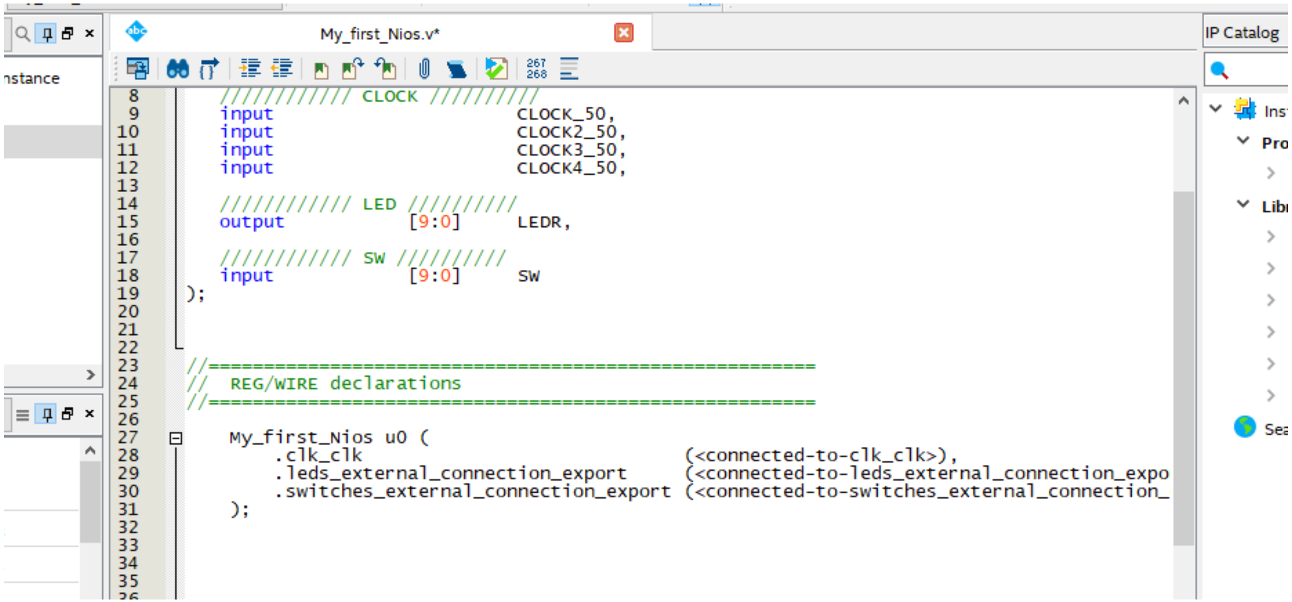
Instanciar el SoPC en Quartus





Instanciar el hardware

Copiar en quartus el template



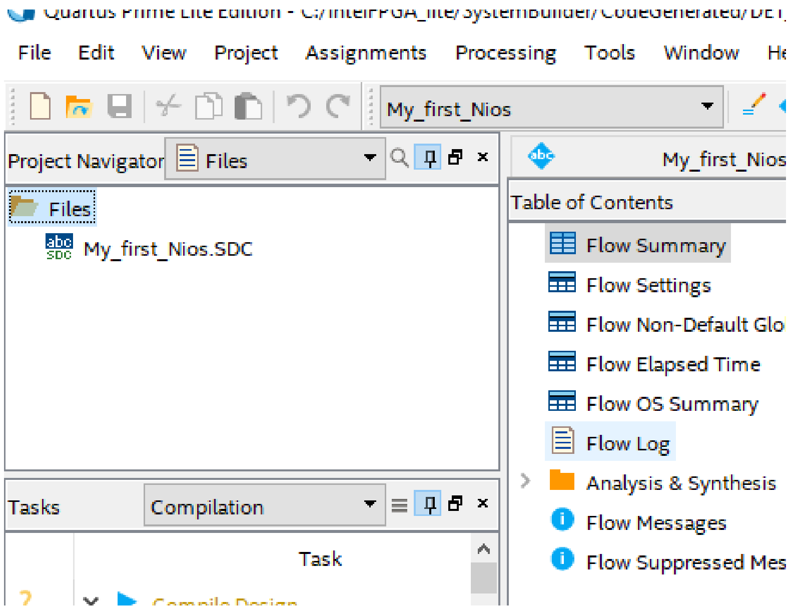
Instanciar el hardware

Asignar las entradas y salidas al nios

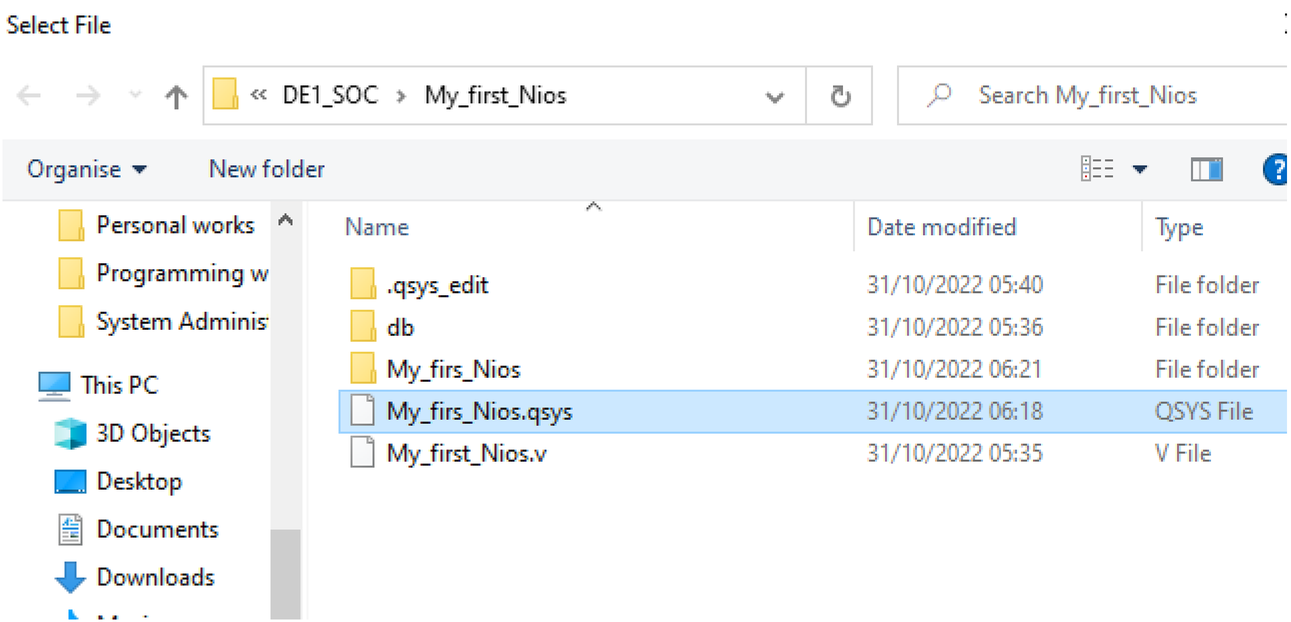
```
    input           [9:0]       SW
);



//=======================================================
//  REG/WIRE declarations
//=======================================================

    My_first_Nios u0 (
        .clk_clk                             (CLOCK_50),
        .leds_external_connection_export     (LEDR[7:0]),      //      leds_external
        .switches_external_connection_export (SW[7:0])  // switches_external_conne
    );






//=======================================================
//  Structural coding
//=======================================================
```

Instanciar el hardware

Antes de compilar el proyecto, ir a la parte de Files y sumar el .qsys generado por Platform Designer



Instanciar el hardware

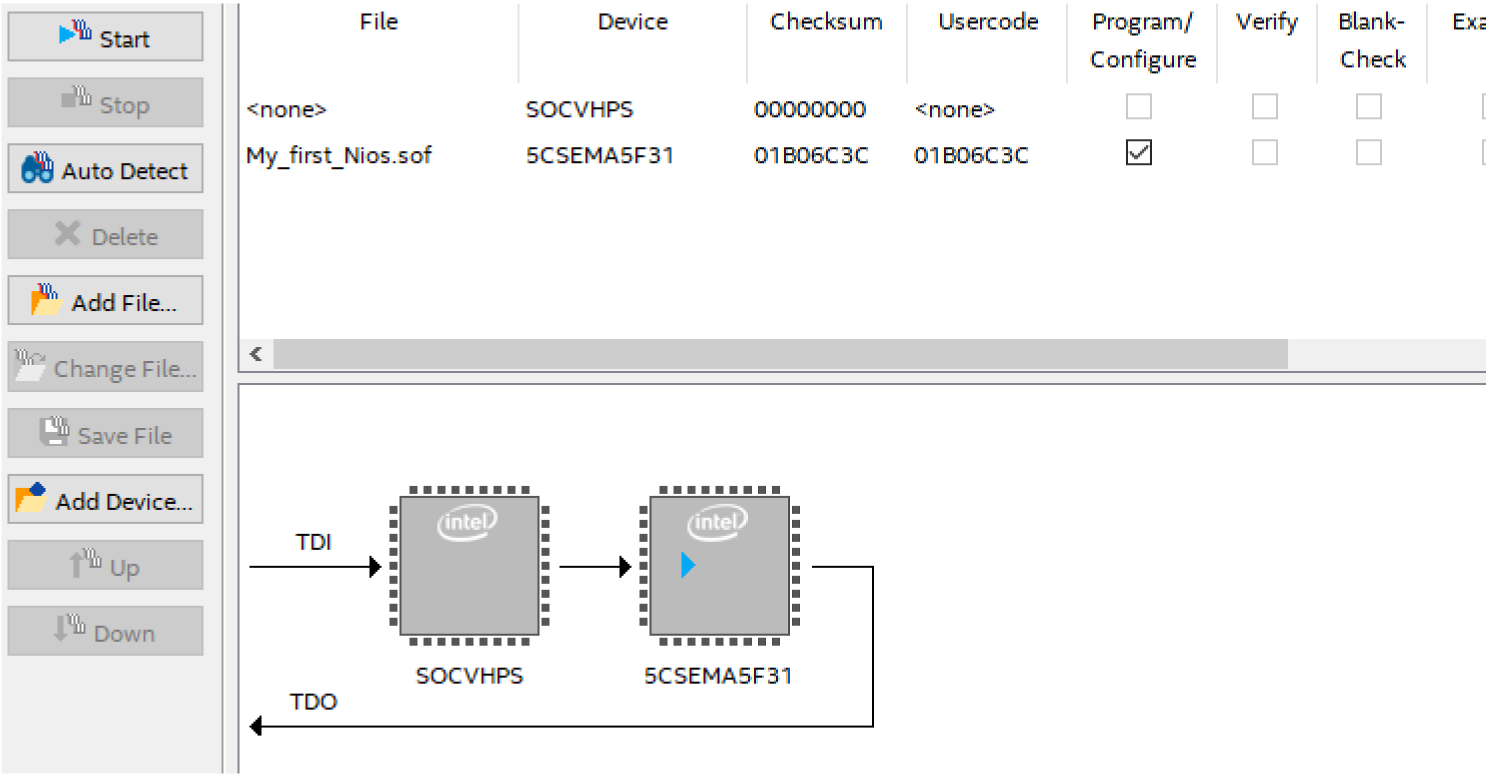


Después de sumarlo al proyecto , compilar

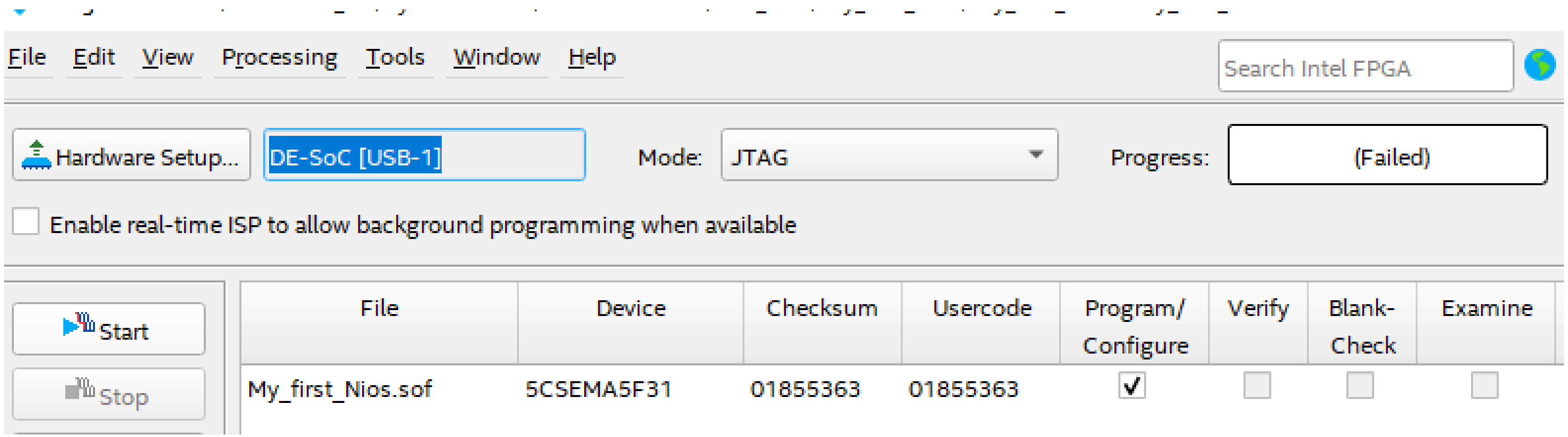
Instanciar el hardware

```
332123 Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer
332146 Worst-case setup slack is 12.535
332146 Worst-case hold slack is 0.009
332146 Worst-case recovery slack is 17.736
332146 Worst-case removal slack is 0.336
332146 Worst-case minimum pulse width slack is 8.438
332114 Report Metastability: Found 9 synchronizer chains.
332102 Design is not fully constrained for setup requirements
332102 Design is not fully constrained for hold requirements
       Quartus Prime Timing Analyzer was successful. 0 errors, 2 warnings
293000 Quartus Prime Full Compilation was successful. 0 errors, 28 warnings
```

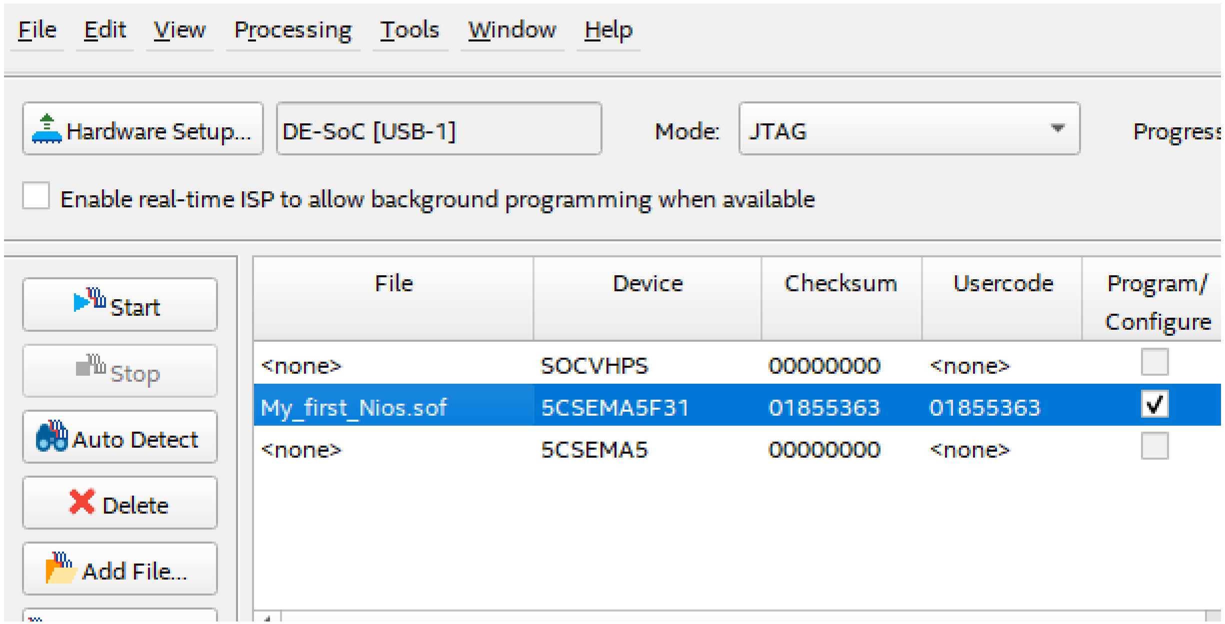
Download hardware a FPGA



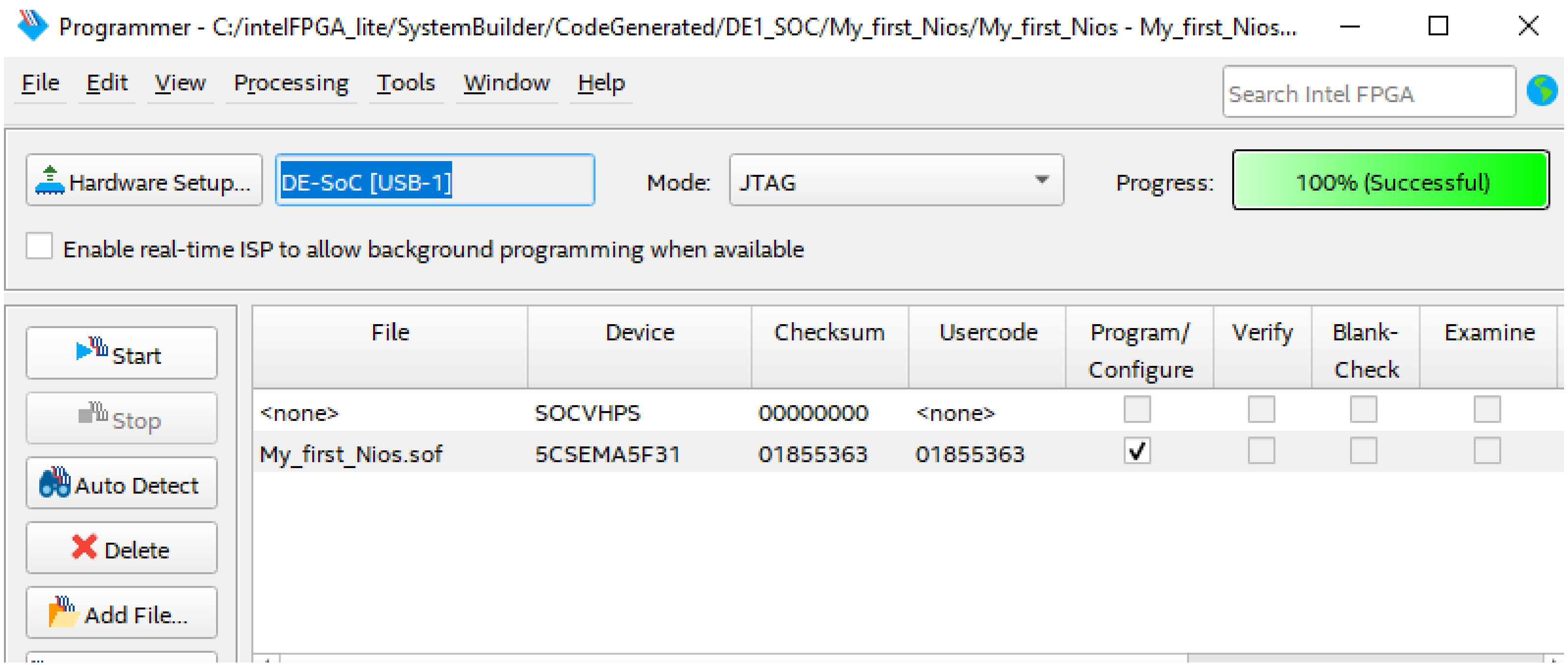
Download hardware a FPGA



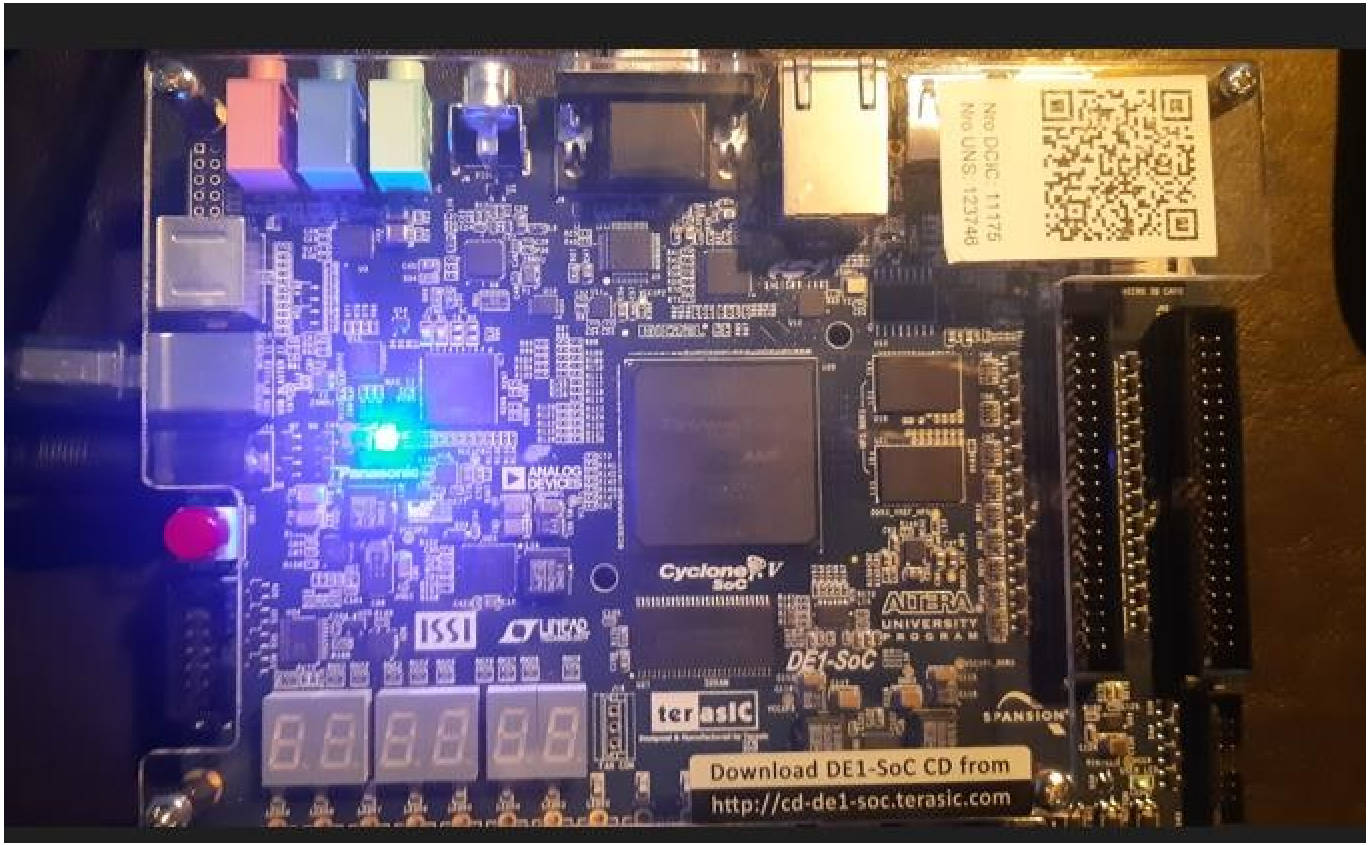
Download hardware a FPGA



Download hardware a FPGA

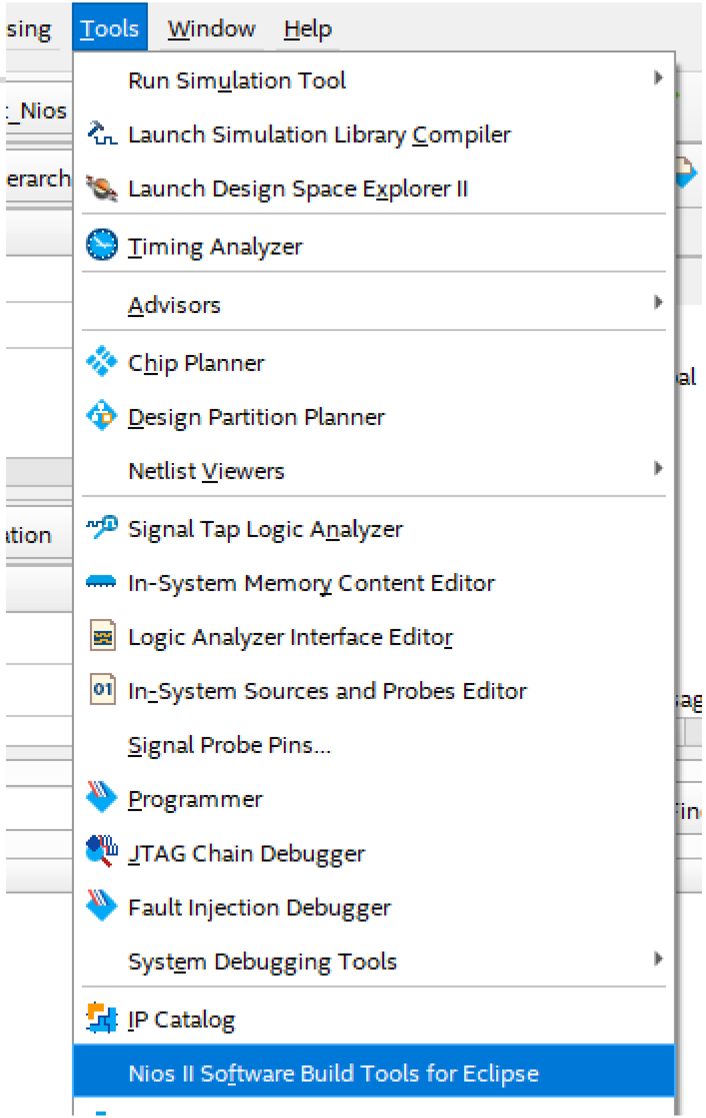


Download hardware a FPGA

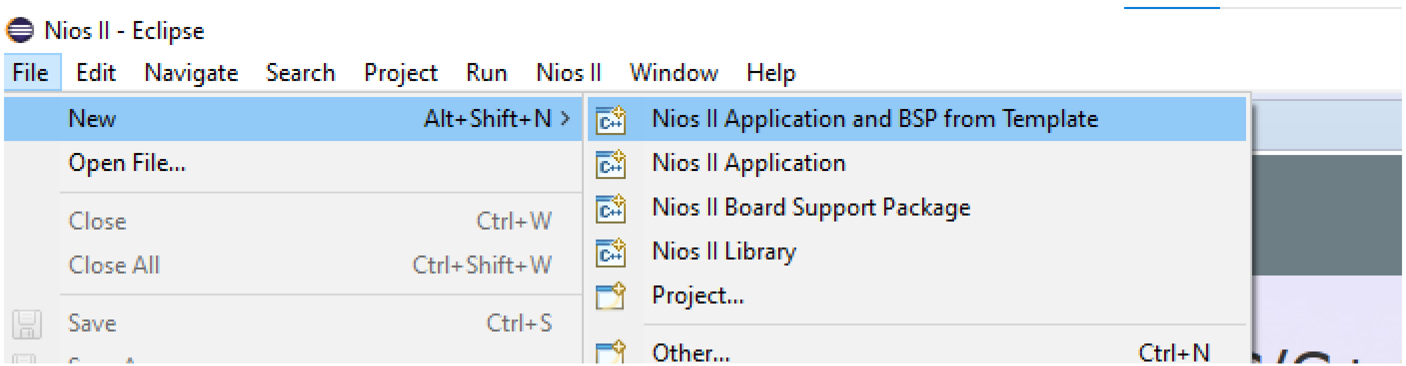


Nios II Software Eclipse

- Crear proyecto para NiosII

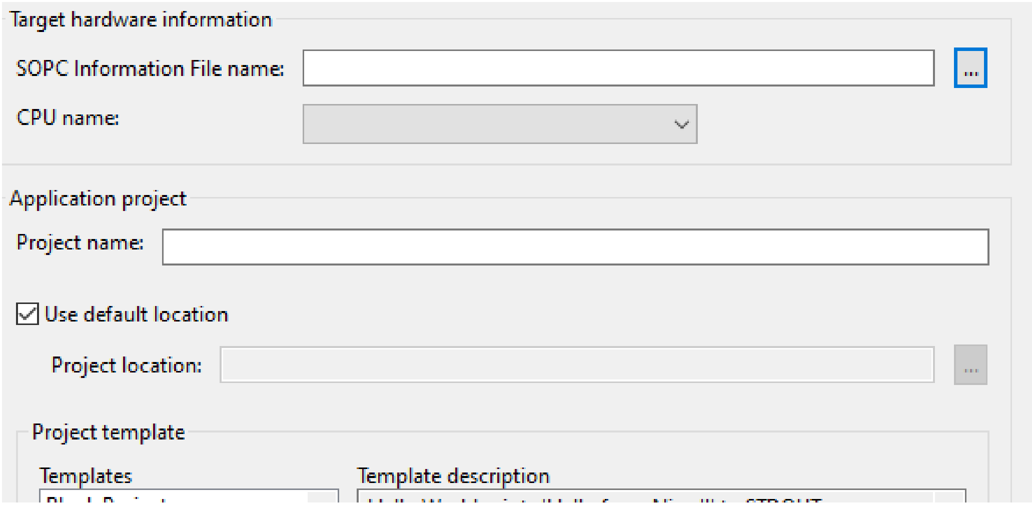


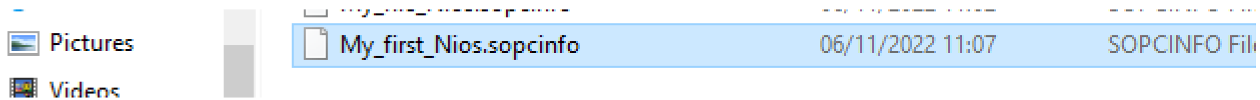
Nios II Software Eclipse



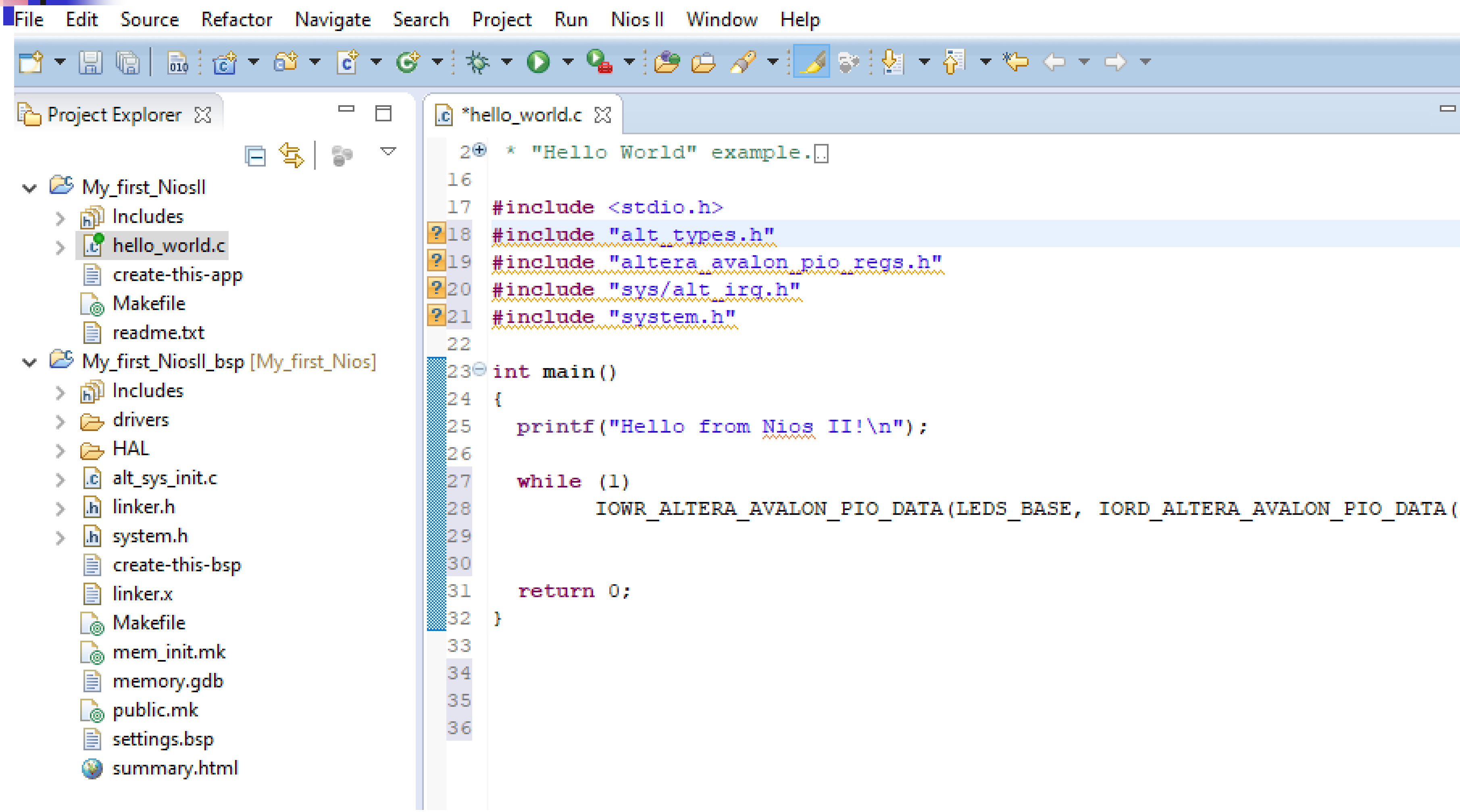
Seleccionar el .sopcinfo generado por el Platform Designer

Nios II Software Eclipse

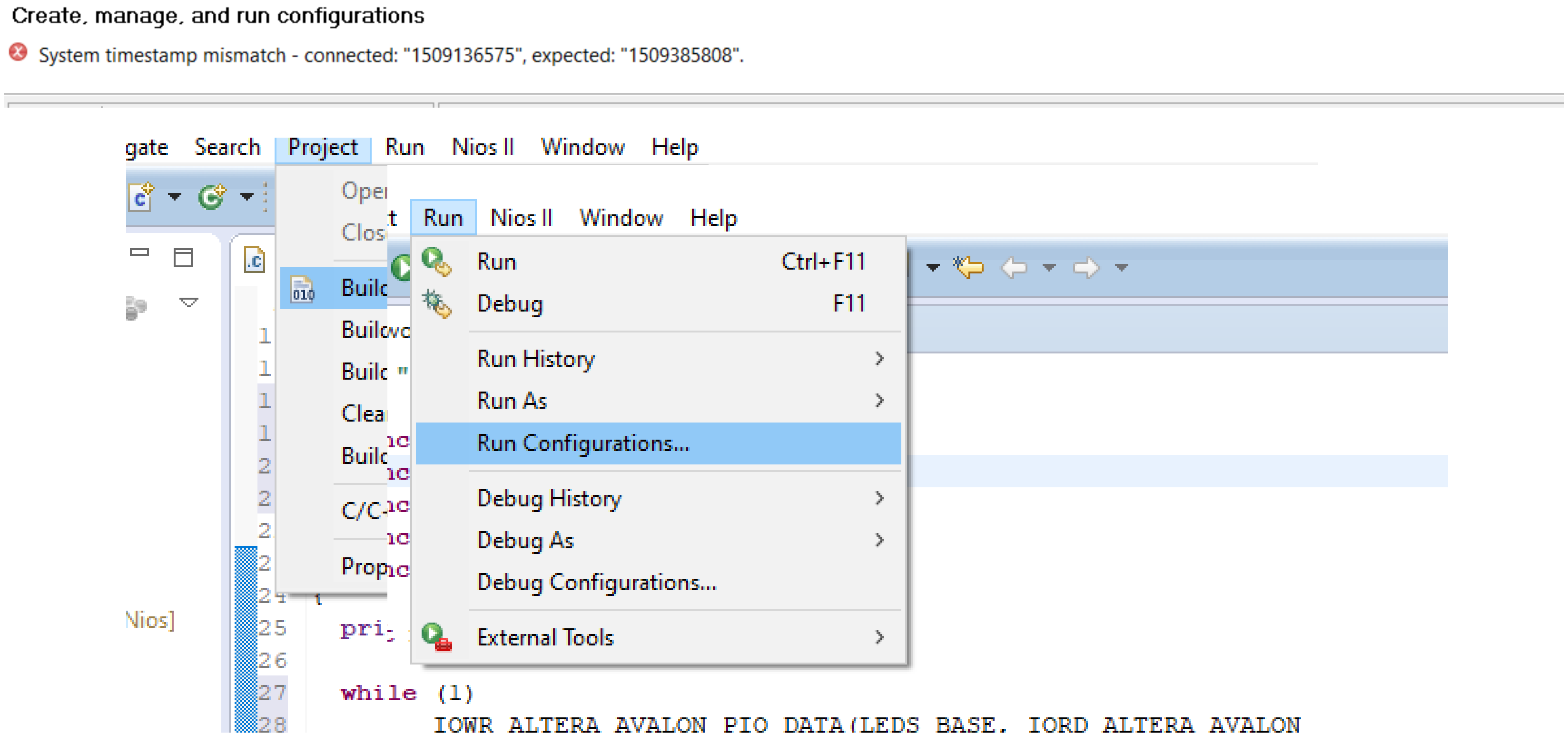




Nios II Software Eclipse



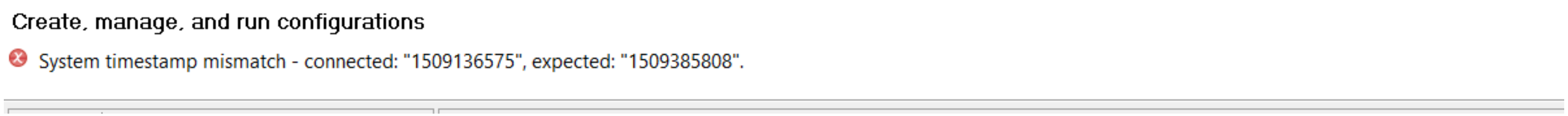
Nios II Software Eclipse

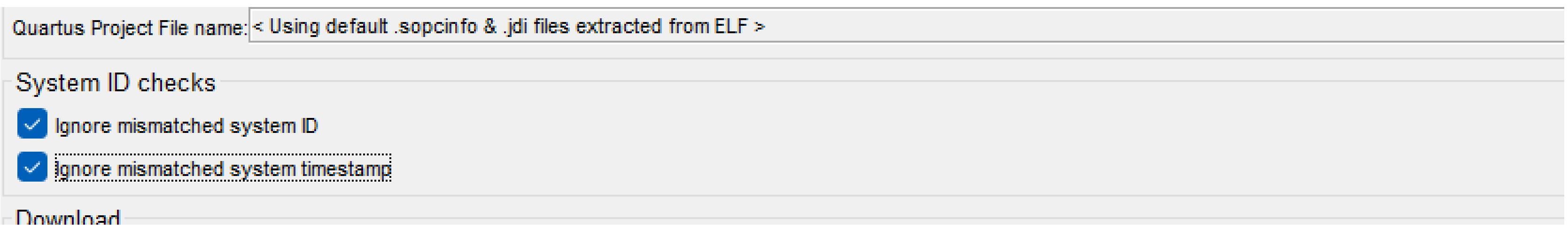


Construir el.elf y volcarlo a la FPGA

Nios II Software Eclipse

- Si obtiene errores de timestamp, setee las opciones de mistmatched





Nios II Software Eclipse

Determinar de dónde el sistema obtiene los valores de LEDS_BASE y SWITCHES_BASE en el main.c

hello_world.c

```c
 * "Hello World" example.

#include <stdio.h>
#include "alt_types.h"
#include "altera_avalon_pio_regs.h"
#include "sys/alt_irq.h"
#include "system.h"

int main()
{
  printf("Hello from Nios II!\n");

  while (1)
        IOWR_ALTERA_AVALON_PIO_DATA(LEDS_BASE, IORD_ALTERA_AVALON_PIO_DATA(SWITCHES_BASE));
  return 0;
}
```

Corroborar el cambio de la entrada (SW) con respecto a la salida (LEDS) en la fpga.

Nios II Software Eclipse